GRADE 3 · INTRO PYTHON TURTLE

Change the Shape

Ontario Math C3.2 — read & alter code

Name: .... Date: ....

★ **I can change a loop to make Bit draw a different shape.**

Same loop, new shape

Bit's square uses range(4) and right 90. To draw a different shape, you change TWO things: how many sides (the range number) and the turn (given to you).

Read the chart — each shape just needs a different range and turn.

Bit's square (start here)

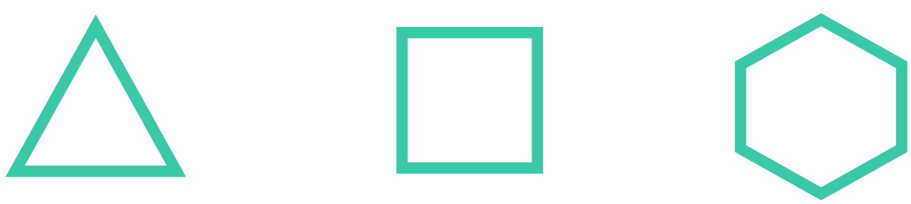
PYTHON

```python
for i in range(4):
    bit.forward(50)
    bit.right(90)
```

OUTPUT

```
A square. Change range and the turn to get other shapes.
```

more sides = bigger range, smaller turn



triangle	**square**	**hexagon**
range(3), right 120	range(4), right 90	range(6), right 60

Change range() and the given turn → triangle, square, or hexagon.

1 Read it

A triangle has 3 sides and the turn is 120 (given). What range number do you use? range(______) and right(______)

____

2 Predict, then run

You change Bit's square to range(6) with right(60). PREDICT: how many sides will the shape have, and what is it called? Then run it.

____

____

3 Write your own

Finish the code so Bit draws a TRIANGLE (the turn 120 is given): for i in range(______): bit.forward(50) bit.right(120)

____

4 Challenge

Bit's loop is range(5) with right(72). How many sides does the shape have, and what is it called? (Hint: count the repeats.)

____

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Big idea

The range number sets how many sides; the turn is given for each shape. Change those two numbers and you change the shape Bit draws.